

USER GUIDE

Settlement Matching and Geocoding Tool

For IM Officers, GIS Specialists, and Data Assistants

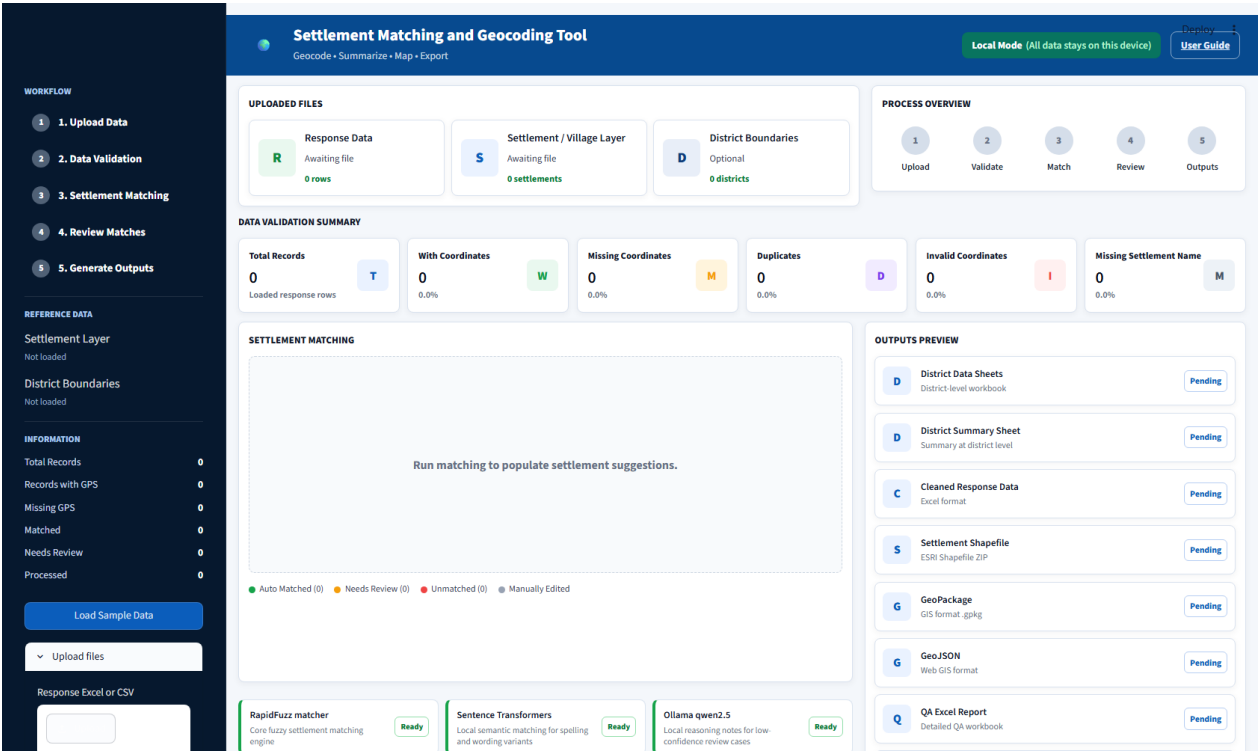
This guide explains how to use the app from a data-operator's point of view: what to prepare, what each screen shows, what the numbers and colors mean, and what to do when a record doesn't match automatically. It assumes no coding knowledge.

● Runs fully offline / Local Mode	No coding required	Streamlit desktop app
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What this tool does

The processor takes a partner's response spreadsheet (settlements visited, beneficiaries reached, etc.), matches any records missing GPS coordinates against a settlement gazetteer, validates the data for common quality problems, and produces cleaned Excel workbooks, GIS files (Shapefile, GeoPackage, GeoJSON), and QA reports — ready to hand off or load into ArcGIS, QGIS, or Power BI.

Everything runs locally on the workstation. No response data, gazetteer, or coordinates are sent to any external server. The header always shows a Local Mode badge as a reminder of this.



Landing screen, before any file is loaded

Before you start: what you need

Input	Required?	Must contain	Notes
Response data	Required	Settlement name, District	Lat/long optional — missing coordinates are what matching fills in. .csv, .xlsx, .xls.
Settlement gazetteer	Required	Settlement, District, Lat, Long	Region recommended. Also accepts a spatial file (.geojson, .gpkg, .zip shapefile).
District boundaries	Optional	—	Adds a boundary overlay to the map. .geojson, .json, .gpkg, .zip.

You don't have to name columns exactly — common humanitarian naming variants are recognized automatically:

- **Settlement** → settlement, village, site, location, town, ...

- **District** → district, admin2, adm2_en, ...
- **Region** → region, admin1, adm1_en, ...
- **Latitude / Longitude** → lat/lon, y/x, gps_latitude/gps_longitude, ...
- Optional fields like **partner**, **cluster/sector**, and **beneficiaries** are also detected and carried into outputs.

If a required column can't be found, Data Validation (Step 2) says so explicitly and names the missing field.

No files yet? Click Load Sample Data in the sidebar for a ready-made example (12 response rows, 11 gazetteer settlements, 5 districts) and try the whole workflow before touching real data.

The five-step workflow

The sidebar tracks progress through five stages — a stage turns green once complete, and the current one is highlighted. The same steps appear as a horizontal tracker at the top of the main panel.

01 Upload Data	02 Data Validation	03 Settlement Matching	04 Review Matches	05 Generate Outputs
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Step 1 — Upload Data

In the sidebar:

- **Load Sample Data** — instantly loads the bundled example dataset.
- **Upload files** (expandable) — three pickers: Response Excel/CSV, Settlement Gazetteer, District Boundary Layer (optional). Choose files, then click Load Uploaded Files.

Once loaded, the Uploaded Files panel confirms what was read: file name, response row count, gazetteer settlement count, boundary district count.

To start over completely, use **Restart Process** in the sidebar — it clears all loaded data, matches, and generated files.

Step 2 — Data Validation

Six at-a-glance metrics appear before matching runs.

Metric	What it means
Total Records	Rows read from the response file.
With Coordinates	Already have valid lat/long — untouched by matching.
Missing Coordinates	No GPS value — these go to settlement matching.
Duplicates	Rows that look like repeats (same settlement, district, partner, cluster). Worth a manual check.
Invalid Coordinates	Lat/long outside valid range (lat -90 to 90, lon -180 to 180) — likely swapped or mistyped.
Missing Settlement Name	No settlement name at all — can't be auto-matched; fix at the source.

The app also checks for districts absent from your gazetteer, and region/district pairs that don't match its hierarchy — both flagged as issues with a red (blocking) or yellow (worth reviewing) severity.

Practical tip: if Missing Settlement Name or Invalid Coordinates is non-zero, fix those in the source spreadsheet and re-upload — no automated step can recover a record with no name or an impossible coordinate.

<<

WORKFLOW

OK

1. Upload Data

2

2. Data Validation

3

3. Settlement Matching

4

4. Review Matches

5

5. Generate Outputs

REFERENCE DATA

Settlement Layer

sample_settlement_gazetteer.csv

District Boundaries

sample_district_boundaries.geojson

INFORMATION

Total Records

12

Records with GPS

5

Missing GPS

6

Matched

0

Needs Review

0

Processed

0

Load Sample Data

Run Matching

Apply Geocodes

Settlement Matching and Geocoding Tool

Geocode • Summarize • Map • Export

Local Mode (All data stays on this device)

Privacy Settings User Guide

UPLOADED FILES

R

Response Data

sample_response.csv

12 rows

S

Settlement / Village Layer

sample_settlement_gazetteer.csv

11 settlements

D

District Boundaries

sample_district_boundaries.geojson

5 districts

PROCESS OVERVIEW

OK

Upload

2

Validate

3

Match

4

Review

5

Outputs

DATA VALIDATION SUMMARY

Total Records

12

Loaded response rows

T

With Coordinates

5

41.7%

W

Missing Coordinates

6

50.0%

M

Duplicates

2

16.7%

D

Invalid Coordinates

1

8.3%

I

Missing Settlement Name

0

0.0%

M

SETTLEMENT MATCHING

Run matching to populate settlement suggestions.

● Auto Matched (0)

● Needs Review (0)

● Unmatched (0)

● Manually Edited

OUTPUTS PREVIEW

D

District Data Sheets

District level workbook

Pending

D

District Summary Sheet

Summary at district level

Pending

C

Cleaned Response Data

Excel format

Pending

S

Settlement Shapefile

ESRI Shapefile ZIP

Pending

G

GeoPackage

GIS format .gpkg

Pending

G

GeoJSON

Web GIS format

Pending

Q

QA Excel Report

Detailed QA workbook

Pending

RapidFuzz matcher

Core fuzzy settlement matching engine

Ready

Sentence Transformers

Local semantic matching for spelling and wording variants

Ready

Ollama qwen2.5

Local reasoning notes for low confidence review cases

Ready

Uploaded files and validation summary, right after loading data

Step 3 — Settlement Matching

Click **Run Settlement Matching** to attempt to geocode every response row missing coordinates.

How matching works, in order

- 1 **Approved alias** — if an analyst has previously confirmed this exact submitted name/district/region maps to a specific gazetteer entry, that confirmed answer is used first.
- 2 **Exact match** — normalized settlement name matches the gazetteer exactly (tried with district+region, then district alone, then region alone, then nationally if the name is unique across the whole gazetteer).
- 3 **RapidFuzz fuzzy match** (always on) — scores the closest gazetteer candidates by text similarity (district-constrained first, then region, then national), never stopping at just the first hit — the top 5 candidates are kept for comparison.
- 4 **Semantic matching** (optional toggle) — a local Sentence Transformers embedding model (all-MiniLM-L6-v2) re-ranks that fuzzy shortlist by meaning, catching names worded differently but meaning the same place. First run downloads the model once.
- 5 **Ollama reasoning notes** (optional toggle) — for “needs review” matches, asks a local Ollama LLM (qwen2.5) for a short plain-English note on which candidate looks most plausible and why — advisory only, it never changes the score or decision itself. Requires Ollama running locally (ollama serve); if unreachable, the app warns and simply skips this step.

Both AI toggles are **off by default** and layer on top of RapidFuzz — never replace it. If a toggle is on but its dependency is missing, the app warns you and continues with RapidFuzz-only matching so a run never fails outright.

Confidence scoring and status

Every candidate is scored on up to six components; if one is unavailable (e.g. no submitted coordinate to check distance against), its weight shifts proportionally to whichever components are available — missing evidence is never scored as good or bad.

Component	Weight
Settlement name similarity	35%
District consistency	20%
Region consistency	10%
Spatial consistency (distance)	15%
Historical evidence (prior approvals)	10%
Semantic similarity	10%

Confidence	Status	What happens
90-100%	Auto Matched	Coordinates applied automatically — unless a hard safeguard below blocks it.
85-89.99%	Needs Review	Flagged for manual follow-up — see Step 4.
0-84.99%	Unmatched	No confident candidate found.

A high score alone is never enough: the app refuses to auto-accept — regardless of confidence — when there's a genuine district/region contradiction, the settlement name exists in more than one district with no way to tell them apart, the candidate is more than 15km from a submitted coordinate, the top two candidates are within 5 points of each other, this exact candidate has been rejected before for this context, or the candidate is missing required district/region data. A blocked match still lands in Needs Review rather than being downgraded further, so nothing is lost — it just isn't auto-applied.

WORKFLOW

OK

1. Upload Data

OK

2. Data Validation

3

3. Settlement Matching

4. Review Matches

5. Generate Outputs

REFERENCE DATA

Settlement Layer
sample_settlement_gazetteer.csv

District Boundaries
sample_district_boundaries.geojson

INFORMATION

Total Records12

Records with GPS5

Missing GPS6

Matched6

Needs Review2

Processed0

Load Sample Data

Run Matching

Apply Geocodes

Beledweyne	Belet Weyne	Beledweyne	100.0%	4.7358	45.2036	Accept
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Auto Matched (4) Needs Review (2) Unmatched (1) Manually Edited

RapidFuzz matcher
Core fuzzy settlement matching engine

Ready

Sentence Transformers
Local semantic matching for spelling and wording variants

Ready

Ollama qwen2.5
Local reasoning notes for low-confidence review cases

Ready

☐ Enable semantic matching with Sentence Transformers ⓘ
Improves settlement matching using local text embeddings. Install requirements or run the project virtual environment if this is unavailable.

☐ Enable Ollama reasoning notes ⓘ
Adds short local AI reasoning notes for low-confidence matches. Start Ollama locally to enable it.

Needs Review Queue — 3 matches awaiting a decision

Accept the suggested match, reject it, or correct the settlement, district, or coordinates directly, then click Save Reviewed Matches below.

Bulk action confidence threshold (%)
85
Check Accept ≥ ThresholdCheck Reject < Threshold

Sort queue by
Confidence (high to low)

Row ID	Submitted Settlement	Submitted District	📍 Suggested Settlement	📍 Suggested District	Confidence %	Latitude
5	Deyniile	Mogadishu	Deeyniile	Mogadishu	93.3	2.0908
4	Dayniile	Mogadishu	Dayniile	Mogadishu	92.0	2.0905
8	Afgoi	Afgooye	Afgooye	Afgooye	82.1	2.1381

Compare Candidates

Inspect the top alternatives the pipeline considered for a specific record, and select a different one if the top suggestion looks wrong.

Record to compare
Row 4: Dayniile

Rank	Settlement	District	Region	Name %	Semantic %	Spatial %	Historical %	Distance (km)	Confidence %	Admin
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GeoPackage
GIS format .gpkg

GeoJSON
Web GIS format

QA Excel Report
Detailed QA workbook

QA / Matching Report
Matched, unmatched, stats

Audit Log (CSV)
Full matching + review audit trail

Audit Log (Excel)
Full matching + review audit trail

Outputs to generate

District Data Sheets xDistrict Summary S... x

Cleaned Response ... xSettlement Shapefile xGeoPackage x

GeoJSON xQA Excel Report xQA / Matching Report x

Audit Log (CSV) xAudit Log (Excel) x

Generate All Outputs

> Individual files

Needs Review queue, Compare Candidates panel, and outputs preview with the audit log

Step 4 — Review Matches

“Needs Review” and “Unmatched” records deserve attention — the app deliberately never auto-applies anything below 90% confidence (or that trips a hard safeguard), so a human always has the final say.

Needs Review Queue

Every needs-review/unmatched row appears in an editable table directly in the Settlement Matching panel. For each row you can:

- **Accept** the suggested match as-is, or **Reject** it.
- **Edit** the suggested settlement, district, or coordinates directly in the table if you already know the right answer.
- Change the **Status** dropdown directly if needed.

Reviewing a large batch? Two tools above the table cut down on row-by-row clicking:

- **Bulk accept/reject by threshold** — drag the confidence slider to a cutoff, then click **Check Accept $\geq$ Threshold** or **Check Reject $<$ Threshold** to tick every matching row at once. Nothing is written until **Save Reviewed Matches** is clicked.
- **Sort queue by** — reorder by confidence (high-to-low or low-to-high) or row ID, to work through the most likely matches first.

Compare Candidates

Below the queue, pick any needs-review record from the dropdown to see the full ranked shortlist the pipeline considered for it side by side — not just the single top guess:

- Settlement, district, region, and every score component (name/semantic/spatial/historical/confidence) for up to 5 candidates.
- Distance in km from a submitted coordinate, when one exists.
- Pick a different candidate from the dropdown and click **Use This Candidate** — this replaces the suggestion, marks the row accepted, and is what gets saved and taught back to the system (not the pipeline's original guess).
- Or click **Reject This Record** to reject it outright without picking an alternate candidate.

Review from the map instead: on the Settlements Preview Map (Section 8), clicking any red or yellow circle jumps straight to that record in this Compare Candidates panel — with the settlement's real-world location still visible while you decide.

When you accept a match (directly, via Compare Candidates, or via a map click), the app remembers it: next time the same settlement name comes up in the same district/region, it's recognized instantly as an approved alias. Rejecting a candidate is remembered too, and repeatedly rejecting the same suggestion makes the app less confident in recommending it again.

- **Still can't resolve a record?** Fix the spelling at the source and re-run matching, turn on semantic matching if you haven't, check the QA Excel Report's “Low Confidence” sheet (Step 5), or source the coordinate manually and re-upload.

Once satisfied, click **Save Reviewed Matches** and then **Apply Geocodes** (sidebar) to write accepted coordinates back into the working dataset used for outputs and the map.

The Settlements Preview Map

A full-width, interactive view of every geocoded record sits below the matching and outputs panels.

- **Base layers** — switch between Light basemap, OpenStreetMap, Satellite imagery, and Topographic via the layer control (top right).
- **Overlays** — toggle district boundaries, settlement response records (clustered, colored by status), review candidates, and a new **submitted coordinates + distance** layer independently.
- **Submitted-to-candidate lines** — for records with an invalid (not merely missing) GPS value, a dashed line connects the original submitted point to the suggested candidate, labeled with the distance.
- **Conflict markers** — a review point gets a thicker dashed ring when it has a flagged administrative or spatial conflict, visible at a glance without opening the row.
- **Click any marker** for settlement name, district, match status, and confidence.
- **Click a red or yellow review-candidate marker** to open that exact record in the Compare Candidates panel (Step 4) — review it with its real-world location still on screen, without hunting for its row in the table.
- **Zoom controls** (top left) plus normal scroll-wheel/drag, like any GIS viewer.
- **Fullscreen button** (top left) — expands the map to fill the browser window.
- **Mini-map** (bottom left) — shows your current viewport in the wider region.

The map updates automatically as you load data, run matching, and apply geocodes — no manual refresh needed.

The interface is divided into several sections:

- Workflow Sidebar (Left):** Contains steps 1-5: 1. Upload Data, 2. Data Validation, 3. Settlement Matching, 4. Review Matches (active), 5. Generate Outputs. Below are sections for Reference Data (Settlement Layer, District Boundaries), Information (Total Records: 12, Records with GPS: 5, Missing GPS: 6, Matched: 6, Needs Review: 2, Processed: 12), and action buttons (Load Sample Data, Run Matching, Apply Geocodes, Generate Outputs, Restart Process).
- Data Table (Top):** A table with columns: Row ID, Submitted Settlement, Submitted District, Suggested Settlement, Suggested District, Confidence %, and Latitud. It shows three rows of data.
- Compare Candidates Panel (Middle):** A form for reviewing a specific record (Row 4: Dayniile). It includes a table of alternatives with columns: Rank, Settlement, District, Region, Name %, Semantic %, Spatial %, Historical %, Distance (km), Confidence %, and Admin. Below the table are buttons for 'Use This Candidate', 'Reject This Record', 'Run Settlement Matching', and 'Save Reviewed Matches'.
- Map View (Bottom):** A map of Ethiopia and Somalia showing settlement locations. A legend on the right lists layers: Light basemap, OpenStreetMap, Satellite imagery, Topographic, District boundaries, Settlement response records, Review candidates, and Submitted coordinates + distance. The map shows several points with numbers (1, 2) and a dashed line connecting two points.

Full-width map with legend, review candidates, and layer control open

Step 5 — Generate Outputs

Use **Outputs to generate** to choose which files you need (all selected by default), then click **Generate All Outputs** (or **Generate Selected Outputs** if narrowed down).

Output	Format	Contents
District Data Sheets	Excel	One sheet per district with response records.
District Summary Sheet	Excel	Aggregated totals by district (and cluster, where available).
Cleaned Response Data	Excel	Full response dataset with applied geocodes and match metadata.
Settlement Shapefile	ESRI Shapefile (zipped)	Point layer for GIS software.
GeoPackage	.gpkg	Same data, modern GIS format.
GeoJSON	.geojson	Web-GIS-friendly point layer.
QA Excel Report	Excel	Readiness metrics, validation issues, full match table, “Low Confidence” sheet, district/cluster summaries.
QA / Matching Report	PDF	Shareable summary of matching results and data quality.
Audit Log (CSV)	CSV	Every score component, the automatic decision, and any human review decision, reviewer, and note — one row per matched record.
Audit Log (Excel)	Excel	Same audit trail as the CSV, in workbook form.

A processing log (settlement_processing_log.txt) and a combined **Download All Outputs** ZIP are generated automatically alongside your selected files. Each output card shows Ready once generated, or an error message if a stage failed.

<<

WORKFLOW

OK1. Upload Data

OK2. Data Validation

OK3. Settlement Matching

OK4. Review Matches

55. Generate Outputs

REFERENCE DATA

Settlement Layer

sample_settlement_gazetteer.csv

District Boundaries

sample_district_boundaries.geojson

INFORMATION

Total Records12

Records with GPS5

Missing GPS6

Matched6

Needs Review2

Processed12

Load Sample Data

Run Matching

Apply Geocodes

Settlement Matching and Geocoding Tool

Geocode • Summarize • Map • Export

Local Mode (All data stays on this device)

Privacy User Guide

UPLOADED FILES

RResponse Data
sample_response.csv
12 rows

SSettlement / Village Layer
sample_settlement_gazetteer.csv
11 settlements

DDistrict Boundaries
sample_district_boundaries.geojson
5 districts

PROCESS OVERVIEW

OKUpload

OKValidate

OKMatch

OKReview

55Outputs

DATA VALIDATION SUMMARY

Total Records
12
Loaded response rows

With Coordinates
5
41.7%

Missing Coordinates
6
50.0%

Duplicates
2
16.7%

Invalid Coordinates
1
8.3%

Missing Settlement Name
0
0.0%

SETTLEMENT MATCHING

Submitted Settlement	Submitted District	Suggested Match	Confidence	Lat	Lon	Action
Goof Gadud	Baidoa	Goof Gadud	100.0%	3.135	43.723	Accept
Daynille	Mogadishu	Daynille	92.0%	2.0905	45.2676	Review
Deynille	Mogadishu	Deeynille	93.3%	2.0908	45.269	Review
Dollow	Doolow	Doolow	100.0%	4.1667	42.0833	Accept
Afgoi	Afgooye	Afgooye	82.1%	2.1381	45.1212	Review
Marka	Marka	Marka	100.0%	1.7159	44.7717	Accept
Beledweyne	Belet Weyne	Beledweyne	100.0%	4.7358	45.2036	Accept

Auto Matched (4)

Needs Review (2)

Unmatched (1)

Manually Edited

OUTPUTS PREVIEW

DDistrict Data Sheets
District level workbook

DDistrict Summary Sheet
Summary at district level

CCleaned Response Data
Excel format

SSettlement Shapefile
ESRI Shapefile ZIP

GGeoPackage
GIS format .gpkg

GGeoJSON
Web GIS format

QQA Excel Report
Detailed QA workbook

RapidFuzz matcher
Core fuzzy settlement matching engine

Sentence Transformers
Local semantic matching for spelling and wording variants

Ollama qwen2.5
Local reasoning notes for low confidence review cases

Outputs generated and ready to download, including the new audit log

Place Intelligence Database: Backup & Import

The app learns as you review: accepted matches become approved aliases, and rejections are remembered too, both stored locally in data/place_intelligence.db. A panel at the bottom of the dashboard lets you manage that history directly.

- **Export Approved Aliases / Review History / Rejected Matches** — CSV downloads of each table.
- **Backup Full Database** — a complete copy of the raw database file.
- **Import** — upload a CSV/Excel alias list or a previous .db backup. Files are validated before anything is written, and imports merge with existing history (increasing approval counts on a repeat match) rather than silently overwriting it.

Hosted deployments: unless the hosting platform provides persistent storage, this database resets on every restart/redeploy. Export a backup regularly if you're relying on accumulated alias history in a hosted instance.

The screenshot displays the Place Intelligence Database application interface. On the left is a dark sidebar with a 'WORKFLOW' section containing five steps: 1. Upload Data (OK), 2. Data Validation (OK), 3. Settlement Matching (OK), 4. Review Matches (active), and 5. Generate Outputs. Below this is 'REFERENCE DATA' with links for 'Settlement Layer' and 'District Boundaries'. An 'INFORMATION' section shows counts for Total Records (12), Records with GPS (5), Missing GPS (6), Matched (6), Needs Review (2), and Processed (12). At the bottom of the sidebar are buttons for 'Load Sample Data', 'Run Matching', 'Apply Geocodes', 'Generate Outputs', and 'Restart Process'. The main area features a map of Kenya and Somalia with several settlement points marked by colored circles (green for 'Auto accepted', red for 'Needs review', orange for 'Unmatched', and blue for 'Already geocoded'). Some points are enclosed in blue boxes with a '2' inside. A legend in the top right corner explains the settlement status and map controls. At the bottom, a 'Place Intelligence Database — Backup & Import' panel contains export buttons for 'Export Approved Aliases', 'Export Review History', 'Export Rejected Matches', and 'Backup Full Database'. It also has an 'Import' section with an 'Upload' button and a note that imports merge with existing history. A footer note states 'Local processing: all data stays on this device.'

Place Intelligence Database panel with export and import controls

Reading the sidebar at a glance

The sidebar's Information panel is a live dashboard of where you stand:

- **Total Records** — rows in the response file.
- **Records with GPS** — already had valid coordinates on upload.
- **Missing GPS** — still need matching.
- **Matched** — auto-matched + needs-review records combined.
- **Needs Review** — subset of Matched not yet at auto-accept confidence.
- **Processed** — rows in the final dataset (outputs/map) after geocodes are applied.

The Reference Data section above it reminds you which gazetteer and boundary files are currently loaded.

Troubleshooting / FAQ

“sentence-transformers is not installed” warning

Semantic matching was on but the package isn't available. The run still completes using RapidFuzz only. Ask whoever manages the installation to run `pip install sentence-transformers`, then try again.

“Ollama is not reachable at localhost:11434” warning

Ollama reasoning notes were on, but no local Ollama server is running. The run still completes without the extra notes. To enable: install Ollama, run `ollama pull qwen2.5`, then `ollama serve`, and re-run matching.

A settlement I know exists isn't matching at all

Check it's actually in the gazetteer (correct spelling, correct district). If genuinely missing, no amount of fuzzy or semantic matching will find it — add it to the gazetteer, or enter the coordinate manually.

The map shows fewer points than my total record count

The map only plots rows with a valid coordinate. Rows still Needs Review or Unmatched won't appear until resolved (Step 4) and geocodes applied.

I want to start over with a different dataset

Click Restart Process in the sidebar — clears loaded files, matches, and generated outputs without reloading the app.

Is any of my data leaving this computer?

No. Core matching (RapidFuzz, Sentence Transformers) runs locally, and Ollama, if used, is also a local process. The header's Local Mode badge is a permanent reminder of this.

A match shows a high score but still landed in Needs Review

One of the hard safeguards tripped — a district/region contradiction, an ambiguous name with no way to disambiguate it, excessive spatial distance, near-tied top candidates, a history of rejection for this exact context, or missing gazetteer metadata. Check Compare Candidates (Step 4) to see why; the safeguard is listed in the row's Reason text.

Quick reference

Range	Status
90-100%	Auto Matched
85-89.99%	Needs Review

Range	Status
0-84.99%	Unmatched

Color	Meaning
Green	Auto-accepted / already geocoded
Amber	Needs review
Red	Unresolved / invalid
Grey	Manually edited / rejected

Dataset	Required fields
Response data	Settlement, District
Gazetteer	Settlement, District, Lat, Long